Transport Layer, Application Layer & Emerging Networking Concepts

Easy-to-understand study notes with networking examples

Functions of Transport Layer

The Transport Layer provides end-to-end communication between applications. Functions: • Segmentation and reassembly of data • Reliable delivery • Flow control • Error control • Multiplexing using port numbers Network Example: A browser sending data to a web server is like dividing a large file into packets, sending them, and reassembling them correctly at the destination.

TCP

TCP (Transmission Control Protocol) is reliable, connection-oriented, and guarantees delivery. 3-Way Handshake: 1. Client → SYN 2. Server → SYN-ACK 3. Client → ACK Real Network Example: Your browser opening a website first establishes a TCP connection. 4-Way Handshake (Connection Termination): 1. FIN 2. ACK 3. FIN 4. ACK Used by HTTP, HTTPS, FTP, SMTP and many other protocols.

UDP

UDP (User Datagram Protocol) is connectionless and faster than TCP. Characteristics: • No connection setup • No delivery guarantee • Low overhead • Very fast Examples: Video streaming, online gaming, DNS queries, voice calls.

Port Addressing

A port identifies a specific application on a device. Examples: HTTP = 80 HTTPS = 443 FTP = 21 SMTP = 25 DNS = 53 Network Example: IP address identifies a building, while port number identifies a room inside that building.

Congestion Control

Leaky Bucket: Packets leave at a fixed rate. Bursty traffic is smoothed. Token Bucket: Tokens accumulate over time. Traffic can be sent when tokens are available. Allows controlled bursts. Network Example: Leaky bucket behaves like water dripping steadily from a bucket. Token bucket behaves like a prepaid card where tokens are saved and spent.

DNS

Domain Name System converts domain names into IP addresses. Example: www.google.com → IP address Network Example: DNS works like a phone contact list. You remember names, not numbers.

HTTP and HTTPS

HTTP transfers web pages. HTTPS is secure HTTP using encryption. Example: Opening a banking website uses HTTPS to protect passwords and financial data.

FTP

File Transfer Protocol transfers files between systems. Example: Uploading website files from a computer to a web hosting server.

SMTP, POP3, IMAP

SMTP sends emails. POP3 downloads emails to a device. IMAP synchronizes emails across multiple devices. Example: Gmail sending mail uses SMTP. Reading mail may use IMAP.

Remote Login: Telnet vs SSH

Telnet: • Not encrypted • Insecure SSH: • Encrypted • Secure Example: Administrators remotely manage Linux servers using SSH.

File Sharing: NFS and SMB

NFS is common in Linux/Unix environments. SMB is common in Windows environments. Example: A shared office folder accessible from multiple computers.

SNMP

Simple Network Management Protocol monitors and manages network devices. Example: Network administrators monitor routers, switches and servers from a central console.

Cloud Networking Basics

Cloud networking provides networking resources through cloud infrastructure. Examples: Virtual networks, cloud servers, load balancers. Benefits: Scalability, flexibility, lower infrastructure cost.

Internet of Things (IoT)

IoT connects physical devices to the internet. Examples: Smart bulbs, smart cameras, smart thermostats, smart watches. Network Example: A smart bulb receives commands through a network and turns on remotely.

MQTT

MQTT is a lightweight publish-subscribe protocol. Components: • Publisher • Broker • Subscriber Example: A temperature sensor publishes readings to a broker. Mobile apps subscribe to receive updates.

CoAP

Constrained Application Protocol is designed for resource-limited IoT devices. Characteristics: • Lightweight • Direct communication • Uses UDP Example: A smart sensor directly communicates with another device on a local network.

Software Defined Networking (SDN)

SDN separates the control plane from the data plane. Components: • SDN Controller • Network Devices Benefits: • Centralized control • Easier management • Better automation Network Example: Instead of configuring every router individually, an SDN controller manages all devices from one place.

TCP vs UDP Comparison

Topic	TCP	UDP
Connection	Connection-Oriented	Connectionless
Reliability	Guaranteed	Not Guaranteed
Speed	Slower	Faster
Examples	HTTP, HTTPS, FTP	Gaming, DNS, Streaming

Important Application Layer Protocols

Protocol	Purpose	Port
DNS	Name Resolution	53
HTTP	Web Browsing	80
HTTPS	Secure Web Browsing	443
FTP	File Transfer	21
SMTP	Email Sending	25
POP3	Email Retrieval	110
IMAP	Email Synchronization	143